0 · Constantes

g = 9,81 m/s² · ρ ≈ 1025 kg/m³ (dulce 1000) · a = H/2 · ω = 2π/T = 2πf · k = 2π/L · c = L/T = ω/k

z = 0 en superficie, −h en el fondo · calculadora en **RAD** · 1 dbar ≈ 1 m

1 · Agua de mar

σ = ρ − 1000 · ρ(S,T,p) = ρ(S,T,0) / [1 − p/K(S,T,p)] · ρ(S,T,0) = A + BS + $CS^{3/2}$ + DS^2

ρ ↑ con S y p, ↓ con T

Mezcla: T = (m₁T₁ + m₂T₂)/(m₁ + m₂) · S = (m₁S₁ + m₂S₂)/(m₁ + m₂) · %A = b/(a+b)·100

Flotabilidad neutra: $m_{lastre} = \rho_{agua} \cdot V - m_{cuerpo}$

2 · Dispersión

$\omega^2 = gk \cdot \tanh(kh)$ · L = (gT²/2π)·tanh(kh) · c = (gT/2π)·tanh(kh) · L/L₀ = c/c₀ = tanh(kh)

Zona	h/L	Dispersión	L	c
Indefinidas	> 1/2	ω² = gk	$L_0 = gT^2/2\pi$ = 1,56T²	gT/2π = 1,56T
Intermedias	1/20–1/2	completa	L₀·tanh(kh)	c₀·tanh(kh)
Reducidas	< 1/20	ω² = gk²h	T√(gh)	√(gh)

k, T → h: **h = arctanh[ω²/(gk)] / k** (ω²/gk ∈ (0,1)) · k, h → T: ω = √(gk·tanh kh)

T, h → k (Newton): $k_{n+1} = k_n - f/f'$ · f = ω² − gk·tanh(kh) · f' = −g·tanh(kh) − ghk·sech²(kh) · $k_0 = \omega^2/g$

Fenton-McKee: $L \approx L_0 \cdot [\tanh((2\pi h/L_0)^{3/4})]^{2/3}$

Guo: $kh \approx (\omega^2 h/g) \cdot [1 - \exp(-(\omega\sqrt{(h/g)})^{5/2})]^{-2/5}$

Peralte ε = H/L (rotura ≈ 1/7) · Ursell U = HL²/h³

Problema de contorno

$\nabla^2\varphi = 0$ · u = ∂φ/∂x, w = ∂φ/∂z · fondo w = 0 · w = ∂η/∂t · ∂φ/∂t + gη = 0 (z = 0)

φ = (gH/2ω)·cosh k(h+z)/cosh kh·sin(kx − ωt) · η = (H/2)·cos(kx − ωt)

3 · Cinemática y presiones (θ = kx − ωt)

u = (H/2)ω·cosh k(h+z)/sinh kh·cos θ · **w** = (H/2)ω·sinh k(h+z)/sinh kh·sin θ

a_x = (H/2)ω²·cosh k(h+z)/sinh kh·sin θ · a_z = −(H/2)ω²·sinh k(h+z)/sinh kh·cos θ

$u_{max,sup}$ = (H/2)ω·coth kh · $u_{max,fondo}$ = (H/2)ω/sinh kh

A = (H/2)·cosh k(h+z)/sinh kh · **B** = (H/2)·sinh k(h+z)/sinh kh · B/A = tanh k(h+z)

Indef.: círculos r = (H/2)e^{kz} · Red.: A = H/(2kh), B = (H/2)(1 + z/h)

$p = -\rho g z + \rho g(H/2) \cdot K_p \cdot \cos\theta$ · K_p = cosh k(h+z)/cosh kh · fondo K_p = 1/cosh kh

Dos sensores: p₁/p₂ = cosh k(h+z₁)/cosh k(h+z₂)

Onda larga: c = √(g·h(x)) · **$t = \int dx/\sqrt{(g \cdot h(x))}$** · talud lineal: t = 2√h₀/(s√g)

4 · Superposición y energía

Estacionaria: η = H_i·cos kx·cos ωt · antinodo $2H_i$, nodo 0 · nodo↔antinodo L/4

Cuasi-estacionaria: η = (H_i/2)cos(kx − ωt) + (H_r/2)cos(kx + ωt + ε)

ancho máx = $H_i + H_r$ · ancho mín = $H_i - H_r$ · máx↔mín = L/4 · $K_r = H_r/H_i$ · $K_t = H_t/H_i$

Crestas cortas: η = 2A·cos(k sinθ·y)·cos(k cosθ·x − ωt)

Grupo: η = H·cos(kx − ωt)·cos[½(Δk·x − Δω·t)] · ω = (ω₁+ω₂)/2 · Δω = ω₂−ω₁ · Δk = k₂−k₁

c = ω/k · c_g = Δω/Δk · $L_g = 2\pi/\Delta k$ · $T_g = 2\pi/\Delta\omega = T_1T_2/(T_1-T_2)$ · $T_{mod} = 4\pi/\Delta\omega$

n = ½[1 + 2kh/sinh 2kh] · c_g = n·c · indef. n = ½ · red. n = 1

$E = \rho g H^2/8$ [J/m²] · $E_c = E_p = \rho g H^2/16$ · **$F = E \cdot c_g$** [W/m] · indef. $P = \rho g^2 H^2 T/32\pi$

Asomeramiento: H₂/H₁ = √(c_{g1}/c_{g2}) · flujo de masa $M = \rho g H^2/(8c)$

Con corriente: $(\omega - kU\cos\theta)^2 = gk \cdot \tanh(kh)$

5 · Estadística

$H_m = \Sigma H_i/N$ · **$H_{1/3}$** = media del 1/3 mayor · **$H_{rms} = \sqrt{(\Sigma H_i^2/N)}$** · $T_m = \Sigma T_i/N$

η normal: $f(\eta) = \exp(-\eta^2/2\eta_{rms}^2) / (\eta_{rms}\sqrt{2\pi})$ · $\eta_{rms} = \sqrt{m_0}$

Rayleigh: $F(H) = 1 - \exp[-(H/H_{rms})^2]$ · $f(H) = (2H/H_{rms}^2)\cdot\exp[-(H/H_{rms})^2]$

$q = \exp[-(H_q/H_{rms})^2]$ · **$H_q = H_{rms}\sqrt{(-\ln q)}$**

Rayleigh	/H_{rms}	Rayleigh	Valor
H_m	0,886	$H_{1/3}/H_{rms}$	**1,416 ≈ √2**
$H_{1/10}$	1,800	$H_{1/3}/\eta_{rms}$	4,004
$H_{1/100}$	2,359	H_{rms}/η_{rms}	2√2 = 2,83
$H_{1/3}/H_m$	1,597	$H_{1/10}/H_{1/3}$	1,271

$F(H_{max}) = \{1 - \exp[-(H_{max}/H_{rms})^2]\}^N$ · **N = duración/T_m**

Moda: $H_{max}/H_{rms} = \sqrt{(\ln N)}$ · **Media: $\sqrt{(\ln N)} + (C/2)(\ln N)^{-1/2}$** · C = 0,5772

N	100	500	1000	2000	5000
media	2,280	2,609	2,738	2,862	3,017
moda	2,146	2,493	2,628	2,757	2,918

Cresta = H/2 · rebase si H/2 > Z_{cub} · $R_u = \eta_{max} + u^2/2g$ · freak: $H/H_s > 2$ ó $\eta_c/H_s > 1{,}25$

6 · Espectral

$m_n = \int \omega^n S(\omega) d\omega$ · m_0 = área = η_{rms}^2 · **$H_{m0} = 4\sqrt{m_0}$** · $E = \rho g m_0$

Parseval: $m_0 = \Sigma a_i^2/2 = \Sigma H_i^2/8$ · mar mixto: $m_0 = m_{0,sea} + m_{0,swell}$ → $H_{m0} = \sqrt{(H_{sea}^2 + H_{swell}^2)}$

Onda regular: $H_{m0} = \sqrt{2}\cdot H$

	S(ω)	S(f)
T_p	$2\pi/\omega_p$	$1/f_p$
$T_{01} = T_m$	2π·m₀/m₁	m₀/m₁
$T_{02} = T_z$	2π·√(m₀/m₂)	√(m₀/m₂)
$T_c = T_{24}$	2π·√(m₂/m₄)	√(m₂/m₄)
$T_e = T_{-10}$	$2\pi \cdot m_{-1}/m_0$	m_{-1}/m_0
T_{ij}	$2\pi(m_i/m_j)^{1/(j-i)}$	sin 2π

$T_c < T_z < T_{01} < T_e < T_p$

$\varepsilon = \sqrt{[1 - m_2^2/(m_0 m_4)]}$ · $\nu = \sqrt{(m_0 m_2/m_1^2 - 1)}$ · $S(f) = 2\pi \cdot S(\omega)$ · $m_n(\omega) = (2\pi)^n m_n(f)$

$P = \rho g \int c_g S\, d\omega$ · indef. $P = \rho g^2 H_{m0}^2 T_e/64\pi \approx 0{,}49 \cdot H_{m0}^2 T_e$ [kW/m]

PM: $S(\omega) = 0{,}0081 g^2 \omega^{-5} \cdot \exp[-1{,}25(\omega/\omega_p)^{-4}]$ · $\omega_p = 0{,}879 g/U_{19,5}$ · $H_{m0} \approx 0{,}04 T_p^2$

JONSWAP: $S_{PM}(\alpha) \cdot \gamma^\delta$ · $\delta = \exp[-(\omega-\omega_p)^2/(2\sigma_0^2\omega_p^2)]$ · γ = 3,3 · σ_0 = 0,07 / 0,09

TMA = JONSWAP·φ(ω,h) · Goda: $S(f) = \beta_J H_s^2 T_p^{-4} f^{-5} \cdot \exp[-1{,}25(T_p f)^{-4}] \cdot \gamma^\delta$

$f_{Nyq} = 1/(2\Delta t)$ · $\Delta f = 1/T_{reg}$ · Welch K tramos: $\Delta f = K/T_{reg}$, error ∝ $1/\sqrt{K}$